Technical Summary: Machine Learning Methodologies

Project: Coronary Artery Segmentation & Stenosis Detection

1. Self-Supervised Representation Learning (Backbone)

Model: facebook/dinov2-small (Hugging Face transformers)

- Extracts high-quality semantic visual features from artery tissue patch inputs.
- Utilizes self-supervised pre-trained weights to capture visual contours, textures, and structures without requiring human-annotated labels.
- Operates with frozen parameters during training to prevent overfitting on the custom dataset.

2. Supervised Segmentation Head (Decoder)

Architecture: 3-layer Convolutional Segmentation Head

- Consists of consecutive 2D convolutions, Batch Normalization, and ReLU activations.
- Decodes the extracted 384-dimensional DINOv2 features into the spatial segmentation maps.
- Outputs pixel-level predictions across the 5 target classes representing vessel anatomy.

3. Anatomy-Aware Loss Functions

Cross-Entropy Loss (CE)

Evaluates pixel-by-pixel classification errors compared to the pseudo-ground truth masks.

clDice (Centerline Dice) Loss

- Applies soft morphological erosion and dilation (soft skeletonization) during backpropagation.
- Enforces connectivity on the Media (outer wall) and Lumen (blood channel) to guarantee biologically plausible loop shapes rather than fragmented boundaries.

4. Unsupervised Data Processing

K-Means Clustering

Groups deep visual feature patches into 5 structural clusters (Background, Media, Intima, Lumen, Extra) to automatically construct ground-truth annotations.

Principal Component Analysis (PCA)

Reduces the 384-dimensional features to 3 principal components mapped as RGB channels to visually confirm feature grouping quality.